1) Aşağıdaki denklemlerden özdeşlik olanlarını belirleyiniz.

- ☐ $x + 2x = 3x$
- ☐ $x + x = x^2$
- ☐ $a + b = ab$
- ☐ $x - y = y - x$
- ☐ $x.x.x.x = x^4$
- ☐ $3.(x + 1) = 3x + 1$
- ☐ $2.(3 + a) = 6 + 2a$
- ☐ $x.(x - y) = x^2 - xy$
- ☐ $(x + 2)^2 = x^2 + 4$
- ☐ $(a - 3)^2 = a^2 - 6a + 9$

2) Aşağıdaki ifadelerin özdeşlerini bulunuz.

a) $(x + 1)^2 =$

b) $(x + 2)^2 =$

c) $(2x + 1)^2 =$

ç) $(3x + 5)^2 =$

d) $(4a + 3b)^2 =$

e) $(y - 1)^2 =$

f) $(a - 2)^2 =$

g) $(7a - 8)^2 =$

ğ) $(5 - x)^2 =$

h) $(9x - 3y)^2 =$

3) Aşağıdaki ifadelerin özdeşlerini iki kare farkı özdeşliğinden faydalanarak bulunuz.

a) $x^2 - 1 =$

b) $y^2 - 4 =$

c) $4z^2 - 9 =$

ç) $36a^2 - 144b^2 =$

d) $16n^2 - 9^2 =$

e) $64 - x^2 =$

f) $(9a)^2 - 25 =$

g) $(2k)^2 - (3p)^2 =$

ğ) $100 - 36x^2 =$

h) $144a^2 - 49b^2 =$

ı) $x^4 - 1 =$

i) $x^8 - x^4 =$

j) $(x + 1)^2 - x^2 =$

k) $(a - b)^2 - (a + b)^2 =$

l) $(a - b + c)^2 - (a - c + b)^2 =$

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4) **Aşağıdaki ifadelerin özdeşlerini bulunuz.**

a) $(x - 3).(x + 3) =$

b) $(2x + y).(2x - y) =$

c) $(3a - 5b).(3a + 5b) =$

ç) $(7 - x).(7 + x) =$

d) $(x + 6).(x + 6) =$

e) $(2n + 4).(2n + 4) =$

f) $(a - 2).(a - 2) =$

g) $(x + 9).(9 - x) =$

5) **Aşağıda istenilenleri özdeşlikler yardımıyla bulunuz.**

$x^2 + y^2 = 7$
$2xy = 18$
olduğuna göre $x + y = ?$

$x^2 + y^2 = 13$
$xy = 2$
olduğuna göre $x - y = ?$

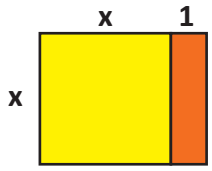
$a + b = 8$
$a.b = 15$
olduğuna göre $a^2 + b^2 = ?$

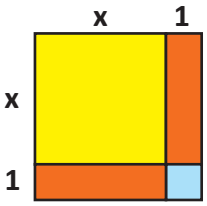
$a + b = 8$
$a - b = 1$
olduğuna göre $a^2 - b^2 = ?$

$x^2 - y^2 = 12$
$x + y = 3$
olduğuna göre $x - y = ?$

$x^2 + y^2 = 29$
$x - y = 7$
olduğuna göre $xy = ?$

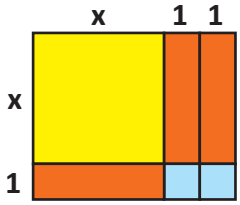
6) **Aşağıda modellenmiş özdeşlikleri yazınız.**

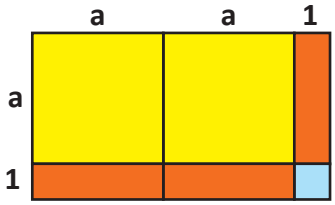




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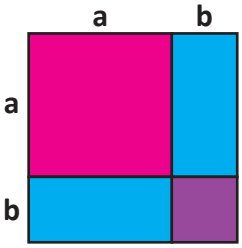
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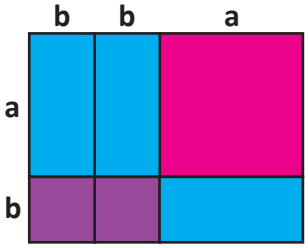




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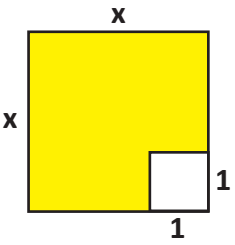


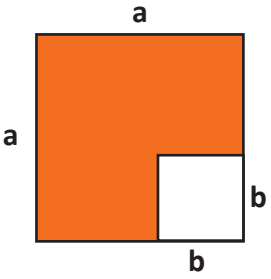


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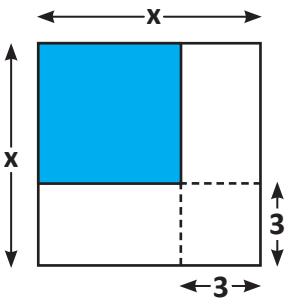
7) **Aşağıdaki taralı alanları temsil eden cebirsel ifadeleri ve bu cebirsel ifadelerin özdeşlerini yazınız.**

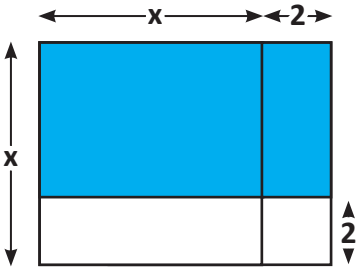




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